



EUCAST guidance on Interpretation of MICs for rare yeast without breakpoints in breakpoint tables

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In the breakpoint tables for antifungal agents, there are species and antifungal agents lacking numerical breakpoints to allow categorical interpretation to S, I or R or a dash to allow the reporting of “resistant” without testing. This is because clinical breakpoint setting requires information on species-specific clinical outcome data, which are generally non-existent for species other than the common pathogens (*C. albicans*, *C. dubliniensis*, *C. glabrata*, *C. krusei*, *C. parapsilosis*, and *C. tropicalis*).

EUCAST recommendations regarding criteria for antifungal agents, which may be considered for use against different fungal species, are given below in Tables 1-2 and are based on the work presented in Astvad *et al* [3]. The classification relies on two important assumptions. First, when isolates are genetically related, pathogenicity and intrinsic susceptibility patterns are assumed to be similar. Second, even if species are not phylogenetically related, the rare yeasts will likely respond to therapy provided the MIC is comparable to values observed in wild-type isolates of more prevalent susceptible species. This second assumption could be explained by the rarity most likely being due to a lower pathogenicity. In addition, the treatment recommendations available in the current guidelines based on the *in vivo* efficacy data and clinical experience were taken into consideration in this classification [1,2].

Importantly, EUCAST recommends against simply reporting an S/I/R interpretation (especially “S” and “I”), as use of these abbreviations may mislead the reader to believe the classification is based on the same extensive amount of data that underlies EUCAST clinical breakpoints. Instead, EUCAST advise to use alternative reporting strategies such as report comments, however this may depend on the local laboratory system (see suggested report comments in Tables 1 and 2). Any alternative reporting should ensure that the guidance is based on available knowledge and not all of the data that is required for formal breakpoint setting. However, R can be used in obvious cases due to high MICs. An example could be as indicated in Tables 1 and 2.

Needless to say, it is important to ascertain that the species identification is correct (e.g. using MALDI-TOF with updated database or sequencing). Plus reproducibility of susceptibility tests may be assessed, by retesting the isolate once or twice to confirm the MIC is stable and representative for the isolate. This is especially required if the MIC is close to the “interpretative values”, where the inherent variation associated with MIC determinations can affect interpretation. Moreover, the adoption of this interpretation guide requires, that the susceptibility test method has been properly validated and found to be in agreement with EUCAST E.Def 7.4 reference MIC testing.

Below, in Table 1, species are grouped beneath antifungal agents and susceptibility. Table 2 lists the susceptibility for the individual species alphabetically, to provide easy interpretation of an isolate in the clinical laboratory.

Changes compared to the 1st version (dated 2024-03-05) are that dashes (“-”) have been replaced by R, *C. dubliniensis* has been deleted as BPs have been set, and the species (*dermatis*) specified in table 2.

Table 1. Overview of interpretation of MICs for rare yeast sorted by susceptibility group.

Amphotericin B	Anidulafungin	Fluconazole	Voriconazole
REPORT: "Formal categorising of the susceptibility of the organism is not possible. A cautious interpretation suggests that the agent may be considered for therapy" when...			
<u>Confirmed MIC ≤1:</u> <i>Candida</i> spp. except <i>C. lusitaniae</i> Rare yeasts <i>Trichosporon</i> spp. (2 nd line)	<u>Confirmed MIC ≤ 0.06:</u> <i>C. inconspicua</i> <i>C. nivariensis</i> <i>C. norvegensis</i> <i>C. pelliculosa</i> <i>C. utilis</i> <i>L. elongisporus</i> <i>P. kluyveri</i> <u>Repeat MIC ≤ 0.125 or</u> <u>absence of <i>fks</i> mutations</u> <i>C. intermedia</i> <i>C. lusitaniae</i> <i>C. palmioleophila</i> <i>C. kefyri</i>	<u>Confirmed MIC ≤2:</u> <i>C. intermedia</i> <i>C. kefyri</i> [MIC ≤1] <i>C. lusitaniae</i> <i>C. metapsilosis</i> <i>C. orthopsilosis</i> <i>C. utilis</i> <i>L. elongisporus</i>	<u>Confirmed MIC ≤0.03:</u> <i>C. intermedia</i> <i>C. kefyri</i> <i>C. lusitaniae</i> <i>C. metapsilosis</i> <i>C. orthopsilosis</i> <i>L. elongisporus</i>
REPORT: "Formal categorising of the susceptibility of the organism is not possible. A cautious interpretation suggests that the agent may be considered for therapy for the following situations: 1) non-severe infection, 2) elevated dose, 3) oral consolidation, 4) no better option" When ...			
	<u>Confirmed MIC 0.125 -0.5:</u> →consider use in some situations (for ex. less severe infections, when no better option is available) <i>C. lipolytica</i> <i>C. magnoliae</i> <i>C. metapsilosis</i> <i>C. orthopsilosis</i> <i>C. pararugosa</i> <i>S. cerevisiae</i> <i>A. adenivorans</i>	<u>Confirmed MIC 2-16:</u> →consider use in some situations (increased dosage and less severe infections) <i>C. fermentati</i> <i>C. nivariensis</i> <i>C. pararugosa</i> <i>C. pelliculosa</i> <i>C. guilliermondii</i> <i>C. palmioleophila</i> <i>C. bovina</i> <i>T. dermatitis</i> (1 st line Alt) <i>Cr. neoformans</i> (2 nd line) <i>S. cerevisiae</i> <i>T. asahii</i> (1 st line Alt)	<u>Confirmed MIC 0.06-0.125:</u> →consider use in some situations (TDM confirmed sufficient exposure, less severe infections or when no better option is available) <i>C. fermentati</i> <i>C. guilliermondii</i> <i>C. lipolytica</i> <i>C. nivariensis</i> <i>C. palmioleophila</i> <i>C. pelliculosa</i> <i>C. utilis</i> <i>S. cerevisiae</i> <i>Cr. neoformans</i> [MIC ≤0.5] <i>T. dermatitis</i> (1 st line)
REPORT: "R" when indicated below. For the others, report: "Formal categorising of the susceptibility of the organism is not possible. An MIC in the indicated range suggests the organism is wild-type, but MICs are higher than for common <i>Candida</i> spp. and there is insufficient data to allow interpretation.			
<u>Confirmed MIC >1:</u> Any isolate →regard resistant	<u>Repeat MIC 0.5-1</u> No evidence that allows recommendation <i>C. fermentati</i> <i>C. guilliermondii</i> <u>Repeat MIC ≥1:</u> →Report as R <i>Cryptococcus</i> <i>Trichosporon</i> , <i>Magnusiomyces</i> , <i>Geotrichum</i> and <i>Rhodotorula</i> (Against due to intrinsic resistance)	<u>Confirmed MIC >16</u> →Report as R <i>C. inconspicua</i> <i>C. lipolytica</i> <i>C. magnoliae</i> <i>C. palmioleophila</i> <i>C. norvegensis</i> <i>P. kluyveri</i> <i>G. candidum</i> <i>R. mucilaginosa</i> <i>M. capitatus</i> <i>M. clavatus</i> <i>A. adenivorans</i>	<u>Confirmed MIC 0.25-1:</u> No evidence that allows recommendations <i>C. inconspicua</i> <i>C. norvegensis</i> <i>P. kluyveri</i> <i>M. capitatus</i> (1 st line Alt) <i>G. candidum</i> (1 st line Alt) <u>Confirmed MIC testing ≥ 2</u> →Report as R <i>A. adenivorans</i> <i>R. mucilaginosa</i> (Against)

Recommendations in parenthesis derive from the clinical guidelines [1,2]. Alt: alternative.

Table 2. Alphabetic species-specific recommendation for EUCAST MIC interpretation for rare yeast without breakpoints. Provided the MIC is confirmed to be below the value indicated the report should be as indicated according to colour below the table, whereas values above should be reported as R.

	Amphotericin B	Anidulafungin	Fluconazole	Voriconazole
<i>Arxula (Blastobotrys)</i>				
<i>A. adenivorans</i> , (<i>B. adenivorans</i>)	≤ 1	≤ 0.5	R	R
<i>Candida</i>				
<i>C. bovina</i>	≤ 1		≤ 16	
<i>C. fermentati</i>	≤ 1	≤ 1?	≤ 16	≤ 0.125
<i>C. guilliermondii</i>	≤ 1	≤ 1?	≤ 16	≤ 0.125
<i>C. inconspicua</i>	≤ 1	≤ 0.06	R	≤ 1?
<i>C. intermedia</i>	≤ 1	≤ 0.125 [^]	≤ 2	≤ 0.03
<i>C. kefyr</i>	≤ 1	≤ 0.125 [^]	≤ 2	≤ 0.03
<i>C. lipolytica</i>	≤ 1	≤ 0.5	R	≤ 0.125
<i>C. lusitaniae</i>	R	≤ 0.125 [^]	≤ 2	≤ 0.03
<i>C. magnoliae</i>	≤ 1	≤ 0.5	R	
<i>C. metapsilosis</i>	≤ 1	≤ 0.5	≤ 2	≤ 0.03
<i>C. nivariensis</i>	≤ 1	≤ 0.06	≤ 16	≤ 0.125
<i>C. norvegensis</i>	≤ 1	≤ 0.06	R	≤ 1?
<i>C. orthopsilosis</i>	≤ 1	≤ 0.5	≤ 2	≤ 0.03
<i>C. palmiophila</i>	≤ 1	≤ 0.125 [^]	≤ 16	≤ 0.125
<i>C. pararugosa</i>	≤ 1	≤ 0.5	≤ 16	
<i>C. pelliculosa</i>	≤ 1	≤ 0.06	≤ 16	≤ 0.125
<i>C. utilis</i>	≤ 1	≤ 0.06	≤ 2	≤ 0.125
<i>Cryptococcus</i>	≤ 1	R		
<i>C. neoformans</i>	≤ 1	R	≤ 16*	≤ 0.5
<i>Geotrichum</i>	≤ 1	R		
<i>G. candidum</i>	≤ 1	R	R	≤ 1? "
<i>L. elongisporus</i>	≤ 1	≤ 0.06	≤ 2	≤ 0.03
<i>Magnusiomyces</i>	≤ 1	R		
<i>M. capitatus</i> , <i>M. clavatus</i>	≤ 1	R	R	≤ 1? "
<i>Pichia kluyveri</i>	≤ 1	≤ 0.06	R	≤ 1?
<i>Rhodotorula</i>	≤ 1	R		
<i>R. mucilaginosa</i>	≤ 1	R	R	R
<i>Saccharomyces cerevisiae</i>	≤ 1	≤ 0.5	≤ 16	≤ 0.125
<i>Trichosporon</i>	R *	R		
<i>T. asahii</i>	R *	R	≤ 16"	
<i>T. dermatis</i>	R *	R	≤ 16'	≤ 0.125 '

[^]Repeat MIC ≤ 0.125 or absence of *fks* mutations

'1st line (guideline recommendation [1,2])

"1st line alternative (guideline recommendation [1,2])

*2nd line therapy (guideline recommendations [1,2])

"?" Formal categorising of the susceptibility of the organism is not possible. An MIC below the indicated value suggests the organism is wild-type, but there is insufficient data allow susceptibility interpretation.

Formal categorising of the susceptibility of the organism is not possible. A cautious interpretation suggests that the agent may be considered for therapy

Formal categorising of the susceptibility of the organism is not possible. A cautious interpretation suggests that the agent may be considered for therapy for the following situations: 1) non-severe infection, 2) elevated dose, 3) oral consolidation, 4) no better option

"R": report as R (resistant).

Reference list

1. Arendrup MC, Boekhout T, Akova M, Meis JF, Cornely OA, Lortholary O. ESCMID and ECMM joint clinical guidelines for the diagnosis and management of rare invasive yeast infections. *Clin Microbiol Infect* [Internet]. 2014 [cited 2014 Oct 27];20 Suppl 3:76–98. Available from: <http://www.ncbi.nlm.nih.gov/pubmed/24102785>
2. Chen SCA, Perfect J, Colombo AL, Cornely OA, Groll AH, Seidel D, et al. Global guideline for the diagnosis and management of rare yeast infections: an initiative of the ECMM in cooperation with ISHAM and ASM. *Lancet Infect Dis*. 2021;21:e375–86.
3. Astvad KMT, Arian-Akdagli S, Arendrup MC. A Pragmatic Approach to Susceptibility Classification of Yeasts without EUCAST Clinical Breakpoints. *J Fungi* [Internet]. 2022;8:141. Available from: <https://www.mdpi.com/2309-608X/8/2/141>